

To what extent can neuroplasticity be harnessed to promote functional recovery following a stroke?

Mechanisms of Recovery and Opportunities for Therapeutic Modulation

Independent Literature-Based Research Project

BSc Biomedical Sciences | University of York

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WORKING DRAFT - APPROX. 10% COMPLETE

Project at a Glance

Research question	To what extent can neuroplasticity be harnessed to promote functional recovery following stroke?
Aim	To critically examine the mechanisms through which neuroplasticity contributes to recovery following stroke and evaluate evidence for interventions designed to promote adaptive neuroplasticity.
Project type	Independent critical literature review / literature-based research project
Primary focus	Adult stroke, neuroplasticity, motor recovery, adaptive vs maladaptive plasticity, rehabilitation and neuroplasticity-modulating interventions

Objectives

- Examine the cellular and systems-level mechanisms underlying neuroplasticity following stroke.
- Evaluate evidence for adaptive, compensatory and maladaptive neuroplasticity during recovery.
- Critically assess interventions designed to modulate neuroplasticity and improve functional recovery.
- Identify limitations in the current evidence and potential future directions, particularly personalised neurorehabilitation.

1. Introduction

-Stroke and neurological recovery

-Defining neuroplasticity

-Neuroplasticity following stroke

-Rationale for the review

-Research question and objectives

2. Methodology

-Literature search strategy

- Plan searches using PubMed and, where available through university access, Web of Science and Scopus.
- Use Google Scholar for supplementary searching and reference chasing.
- Initial search concepts: “neuroplasticity” AND “stroke”; “neuroplasticity” AND “motor recovery”; “adaptive plasticity” AND stroke; “maladaptive plasticity” AND stroke; “neural reorganisation” AND rehabilitation.

-Inclusion and exclusion criteria

- Define eligibility by population, topic relevance, study type, publication date and language.

- Include both mechanistic research and clinical evidence where they answer different parts of the research question.

-Evidence evaluation

- Compare study design, sample size, methodology, outcome measures, strengths, limitations and relevance.
- Distinguish mechanistic evidence from evidence of clinical effectiveness.

3. Mechanisms of Neuroplasticity Following Stroke

-Synaptic plasticity

-Structural plasticity

-Cortical reorganisation

-Network-level plasticity

4. Adaptive, Compensatory and Maladaptive Neuroplasticity

-Adaptive neuroplasticity

- Examine evidence that neural reorganisation contributes to improved function.

-Compensatory mechanisms

- Explore recovery achieved through alternative neural pathways or movement strategies rather than complete restoration of the original system.

-Maladaptive neuroplasticity

- Investigate evidence for plastic changes that are inefficient, unhelpful or potentially interfere with recovery.

-Factors influencing plasticity

- Consider stroke severity, lesion location, timing, age, baseline function, rehabilitation dose and integrity of key neural pathways.

5. Therapeutic Modulation of Neuroplasticity

-Task-specific rehabilitation

- Investigate how repeated, goal-directed practice may influence neural reorganisation and motor recovery.

-Constraint-induced movement therapy

- Examine evidence that encouraging use of the affected limb can promote functional recovery and adaptive plasticity.

-Non-invasive brain stimulation

- Evaluate TMS and tDCS, distinguishing changes in neural activity from evidence of meaningful functional benefit.

-Emerging approaches

- Potential areas include vagus nerve stimulation, robotics, virtual reality and brain-computer interfaces.
- Assess the maturity of evidence and translational limitations for each approach.

6. Critical Evaluation of the Evidence

-Does neural change equal functional recovery?

- Examine whether increased activity/connectivity demonstrates causation or may simply accompany behavioural improvement.

-Restoration versus compensation

- Assess whether functional gains reflect recovery of damaged systems, alternative strategies, or a combination.

-Patient heterogeneity

- Investigate why patients respond differently to similar interventions and which clinical or biological factors may explain this.

-Evidence limitations

- Compare animal and human evidence, small mechanistic studies and clinical trials, short-term and long-term outcomes, and biomarkers versus meaningful functional outcomes.

7. Future Directions

-Personalised neurorehabilitation

- Explore whether treatment can be tailored to individual lesion characteristics, baseline function and biomarkers.

-Improved measurement of neuroplasticity

- Consider the strengths and limitations of fMRI, EEG, TMS, neuroimaging and other biomarkers.

-Combination therapies

- Investigate whether rehabilitation combined with neuromodulation or emerging technologies may offer greater benefit than individual approaches.

-Translational research gaps

- Identify what is still needed to move promising mechanistic findings into reliable clinical treatments.

8. Conclusion

-Final synthesis

- Answer the research question based on the overall evidence.
- Summarise what is well established, what remains uncertain and which research questions are most important going forward.

Potential Figures and Tables

Item	Working title	What it should show
Figure 1	Overview of neuroplasticity following stroke	Stroke → neural injury → neural reorganisation → adaptive/compensatory/maladaptive outcomes.
Figure 2	Molecular mechanisms of synaptic plasticity	Potential pathway involving NMDA/AMPA receptors, calcium signalling and downstream plasticity mechanisms.
Figure 3	Adaptive vs compensatory vs maladaptive plasticity	Conceptual comparison showing how different forms of reorganisation may affect function.
Figure 4	Therapeutic approaches targeting neuroplasticity	Rehabilitation, constraint-induced therapy, neuromodulation and emerging technologies.
Table 1	Key evidence summary	Study design, population/model, intervention, principal finding, limitation and relevance.
Table 2	Therapeutic approaches	Intervention, proposed mechanism, clinical evidence, limitations and research gaps.

Evidence Extraction Table - Working Template

Paper / year	Study type	Population / model	Intervention / exposure	Main finding	Key limitation	Relevance to research question
[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]
[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]
[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]
[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]
[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]	[Add evidence]

Literature Search Log - Working Template

Date	Database	Search string	Filters	Results retrieved	Notes / papers to screen
[date]	[database]	[search string]	[filters]	[n]	[notes]
[date]	[database]	[search string]	[filters]	[n]	[notes]
[date]	[database]	[search string]	[filters]	[n]	[notes]
[date]	[database]	[search string]	[filters]	[n]	[notes]
[date]	[database]	[search string]	[filters]	[n]	[notes]
[date]	[database]	[search string]	[filters]	[n]	[notes]